

Sample Mathematics Admissions Paper

Paper 1

Time allowed: 40 minutes

There are four questions in this paper. Attempt all of them.

Calculators and formula booklets are NOT permitted.

1

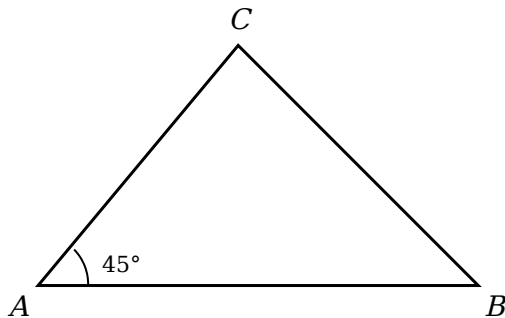
Find the complete set of values of x for which

$$\frac{x^2 - 4}{x + 1} \leq 0$$

- A $x \leq -2$
- B $-2 \leq x < -1$ or $x \geq 2$
- C $x \leq -2$ or $-1 < x \leq 2$
- D $-1 < x \leq 2$
- E $x \geq 2$

2

The diagram shows triangle ABC with $AB = 9/\sqrt{2}$ and angle $BAC = 45^\circ$.

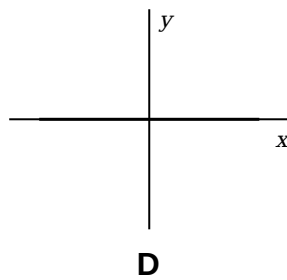
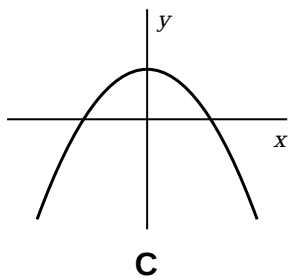
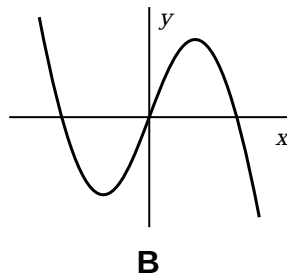
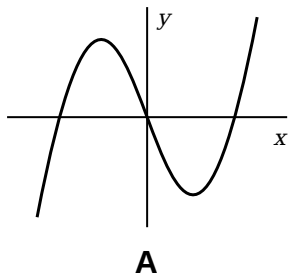


What is the area of triangle ABC ?

- A $\frac{81}{4}$
- B $\frac{81}{2}$
- C $\frac{81\sqrt{2}}{4}$
- D 81

3

Which one of the following graphs could be the graph of $y = x^3 - x$?



- A** Graph A
- B** Graph B
- C** Graph C
- D** Graph D

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4

Evaluate

$$\sum_{k=1}^{10} \log_2 \left(\frac{k+1}{k} \right)$$

A $\log_2 11$

B $\log_2 10$

C $1 + \log_2 11$

D 11

E $\log_{10} 2$